RESPIRATORY ADJUSTMENTS IN ALTERED STATES

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LECTURER



LEARNING OBJECTIVES:

By the end of lecture students should be able to:

- **Explain** how exercise increases pulmonary ventilation, oxygen diffusion, and consumption to meet metabolic demands, creating an oxygen debt that must be repaid post-workout.
- **Define** VO2 max as the absolute limit of oxygen utilization and Respiratory Quotient as the marker of metabolic fuel shifts and lactic acid buffering.
- **Explain** how high altitude environments decrease barometric and oxygen partial pressures, inducing hypoxia, gas expansion, extreme cold, and radiation effects that trigger mountain sickness or necessitate physiological acclimatization.
- **Define** aviation and space physiology principles, specifically focusing on the cardiovascular impact of gravitational (G) forces, protective countermeasures, and the systemic effects of spacecraft travel.
- **Explain** how hyperbaric environments increase barometric pressure with depth, causing nitrogen narcosis through increased gas solubility and decompression sickness when rapid ascent forms bubbles in tissues.
- **Define** the clinical presentations, preventive strategies, treatment protocols (like hyperbaric oxygen), associated with these deep-sea physiological challenges.

RESPIRATORY ADAPTATIONS:

- **Respiratory adaptations** refer to the changes your respiratory system makes to keep your body's oxygen levels and acid-base balance stable (homeostasis) under changing conditions.
- Respiratory adaptation refers to the immediate (acute) adjustments and long-term (chronic) structural modifications made by the respiratory system to optimize oxygen (O_2) uptake and carbon dioxide (CO_2) elimination when faced with metabolic or environmental stress.

ACUTE VS CHRONIC TIMELINES:

- Acute (Immediate): Functional changes occurring within seconds to minutes (e.g., hyperpnea during a sprint).
- Chronic (Long-term): Structural transformations requiring weeks to years (e.g., increased capillary density around alveoli).
- Primary Systemic Triggers: Driven by changes in tissue metabolic demand (exercise), alterations in atmospheric partial pressure (high-altitude hypoxia), or internal chemical shifts (metabolic acidosis).

EFFECTS OF EXERCISE ON RESPIRATION:

EFFECTS ON PULMONARY VENTILATION:

- Pulmonary ventilation is the amount of air that enters and leaves the lungs in 1 minute. It is the product of tidal volume and respiratory rate. It is about 6 liter/minute, with a normal tidal volume of 500 mL and respiratory rate of 12/minute.
- During exercise, hyperventilation, which includes increase in rate and force of respiration occurs. In moderate exercise, respiratory rate increases to about 30/minute and tidal volume increases to about 2,000 mL. Thus, the pulmonary ventilation increases to about 60 L/ minute during moderate exercise. In severe muscular exercise, it rises still further up to 100 L/minute.

FACTORS INCREASING PULMONARY VENTILATION DURING EXERCISE:

1. Higher centers
2. Chemoreceptors
3. Proprioceptors
4. Body temperature
5. Acidosis.

1. Higher Centers:

- Rate and depth of respiration increase during the onset of exercise. Sometimes, before starting the exercise, thought or anticipation of exercise itself increases the rate and force of respiration.
- It is a psychic phenomenon due to the activation of higher centers like Sylvian cortex and motor cortex of brain. Higher centers, in turn accelerate the respiratory processes by stimulating respiratory centers.

2. Chemoreceptor:

Chemoreceptors which are stimulated by exercise induced hypoxia and hypercapnea, send impulses to the respiratory centers. Respiratory centers, in turn increase the rate and force of respiration.

3. Proprioceptors:

Proprioceptors, which are activated during exercise, send impulses to cerebral cortex through the somatic afferent nerves. Cerebral cortex, in turn causes hyperventilation by sending impulses to the medullary respiratory centers.

4. Body Temperature:

Body temperature which increases by muscular activity, increases the ventilation by stimulating the respiratory centers

5. Acidosis:

Acidosis developed during exercise also stimulates the respiratory centers, resulting in hyperventilation.

EFFECT ON DIFFUSING CAPACITY FOR OXYGEN:

- Diffusing capacity for oxygen is about 21 mL/minute at resting condition. It rises to 45 to 50 mL/minute during moderate exercise because of increased blood flow through pulmonary capillaries.

EFFECT ON CONSUMPTION OF OXYGEN:

- Oxygen consumed by the tissues, particularly the skeletal muscles is greatly enhanced during exercise. Because of vasodilation in muscles during exercise, more amount of blood flows through the muscles and more amount of oxygen diffuses into the muscles from blood. The amount of oxygen utilized by the muscles is directly proportional to the amount of oxygen available.

EFFECT ON OXYGEN DEBT:

- Oxygen debt is the extra amount of oxygen required by the muscles during recovery from severe muscular exercise. After a period of severe muscular exercise, amount of oxygen consumed is greatly increased. Oxygen required is more than the quantity available to the muscle. This much of oxygen is required not only for the activity of the muscle but also for reversal of some metabolic processes such as:

1. Reformation of glucose from lactic acid, accumulated during exercise

2. Resynthesis of ATP and creatine phosphate

3. Restoration of amount of oxygen dissociates from hemoglobin and myoglobin.

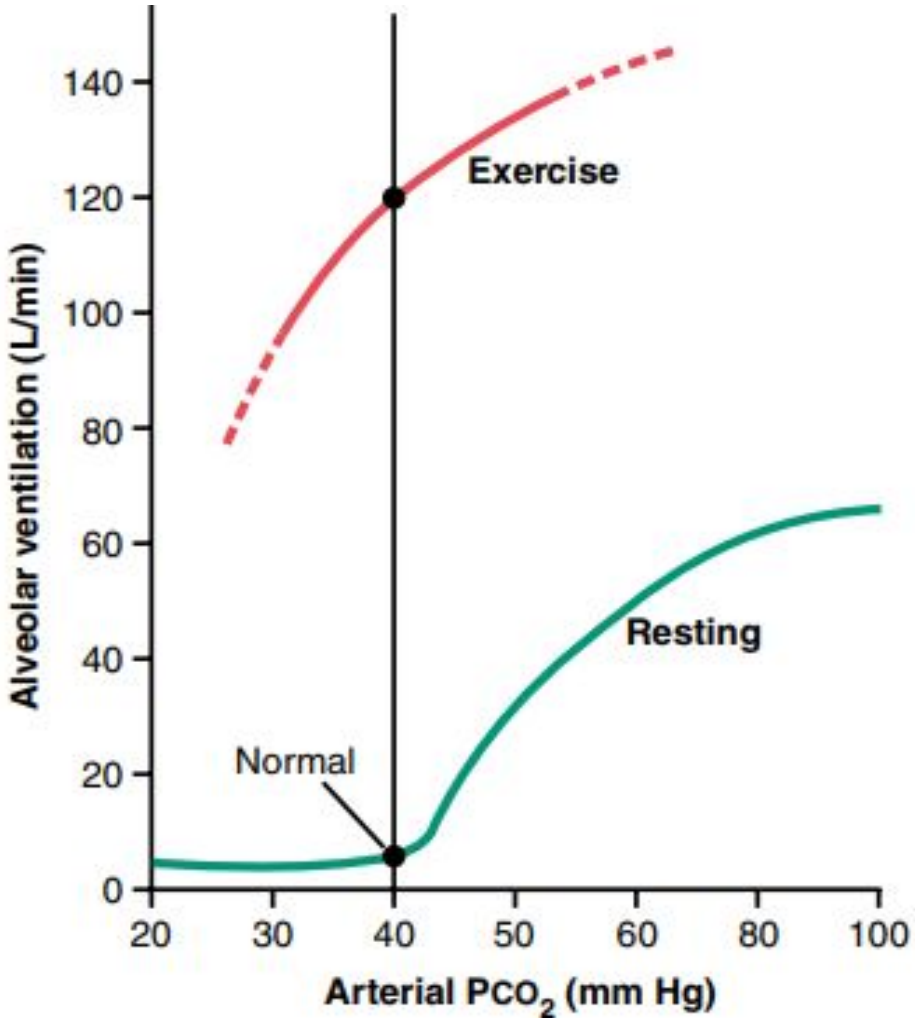
Thus, for the above reversal phenomena, an extra amount of oxygen must be made available in the body after severe muscular exercise. Oxygen debt is about six times more than the amount of oxygen consumed under resting conditions.

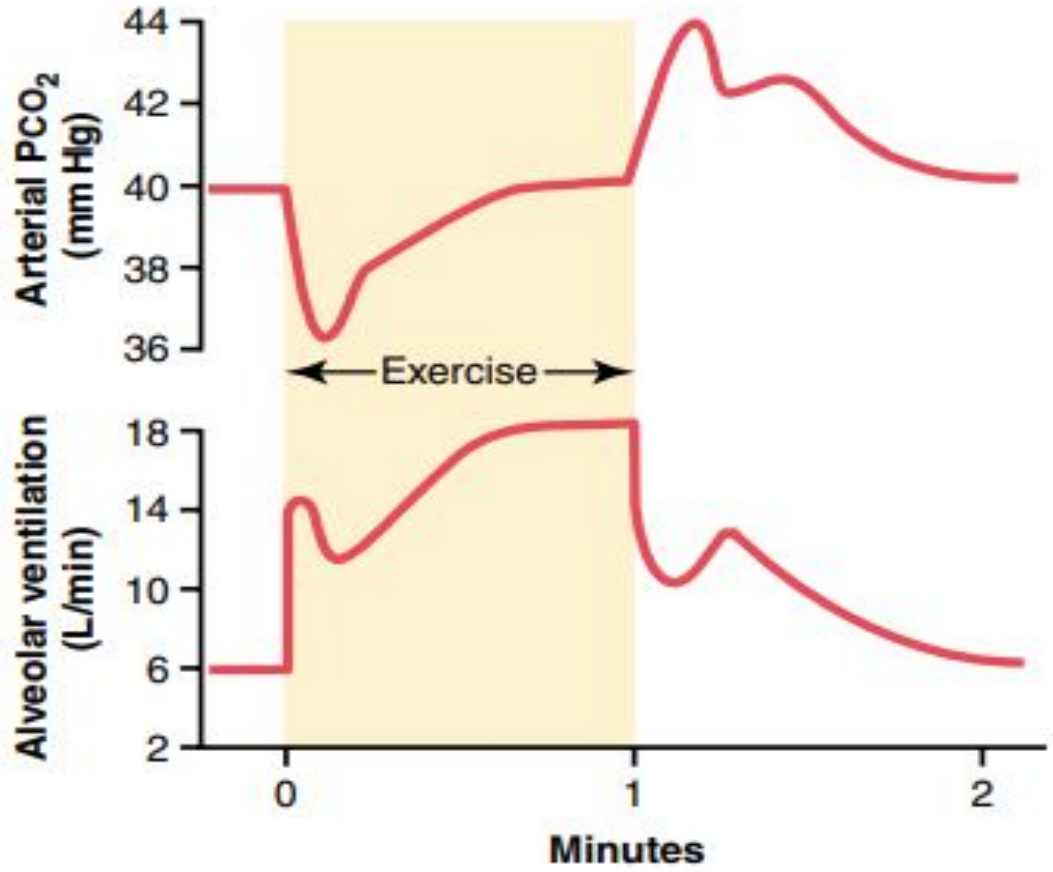
EFFECT ON VO2 MAX:

- VO2 max is the amount of oxygen consumed under maximal aerobic metabolism. It is the product of maximal cardiac output and maximal amount of oxygen consumed by the muscle. In a normal active and healthy male, the VO2 max is 35 to 40 mL/kg body weight/minute. In females, it is 30 to 35 mL/kg body weight/minute. During exercise, VO2 max increases by 50%.
- EFFECT ON RESPIRATORY QUOTIENT: Respiratory quotient is the molar ratio of carbon dioxide production to oxygen consumption. Respiratory quotient in resting condition is 1.0 and during exercise it increases to 1.5 to 2. However, at the end of exercise, the respiratory quotient reduces to 0.5.

INTERRELATION BETWEEN CHEMICAL FACTORS AND NERVOUS FACTORS IN THE CONTROL OF RESPIRATION DURING EXERCISE:

- **Exercise begins:** The brain sends anticipatory nervous signals to the respiratory center.
- **Ventilation increases:** The respiratory center increases alveolar ventilation before CO_2 accumulates in the blood.
- **CO_2 production increases:** Active muscles produce more CO_2 during exercise.
- **Chemical adjustment:** Increased CO_2 stimulates respiration if ventilation is insufficient; if ventilation is excessive, reduced CO_2 decreases respiratory stimulation.
- **Normal PCO_2 maintained:** After 30–40 seconds, ventilation matches CO_2 production, maintaining arterial PCO_2 near **40 mmHg**.


Alveolar ventilation (L/min)
140
120
100
80
60
40
20
0
Exercise
Resting
Normal
20
30
40
50
60
80
100
Arterial PCO_2 (mm Hg)


Arterial PCO_2 (mm Hg)
44
42
40
38
36
Exercise
Alveolar ventilation (L/min)
18
14
10
6
2
0
1
2
Minutes

EFFECT ON HIGH ALTITUDE AND SPACE PHYSIOLOGY:

HIGH ALTITUDE:

- Characteristic feature of high altitude is the low barometric pressure. However, amount of oxygen available in the atmosphere is same as that of sea level. Due to low barometric pressure, partial pressure of gases, particularly oxygen proportionally decreases. It leads to hypoxia. Carbon dioxide in high altitude is very much negligible and it does not create any problem

Altitude	Feet	Meter
High altitude	8,000 to 13,000	2,500 to 4,000
Very high altitude	13,000 to 18,000	4,000 to 5,500
Extreme altitude	> 18,000	> 5,500

BAROMETRIC PRESSURE AND PARTIAL PRESSURE OF OXYGEN AT DIFFERENT ALTITUDES:

- Barometric pressure decreases at different altitudes. Accordingly, partial pressure of oxygen also decreases and produces various effects on the body.

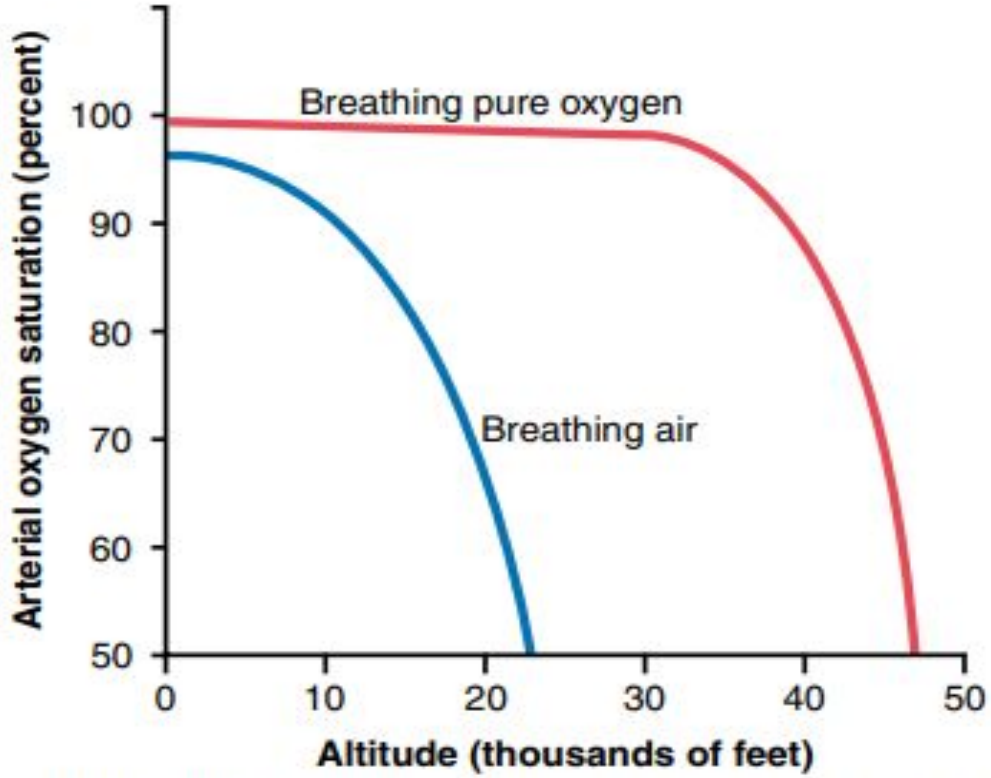


Figure 43-1 Effect of high altitude on arterial oxygen saturation when breathing air and when breathing pure oxygen.

TABLE 128.2: Barometric pressure, partial pressure of oxygen and common effects at different altitudes

Altitude (feet)	Barometric pressure (mm Hg)	Partial pressure of oxygen (mm Hg)	Common effects
Sea level	760	159	–
5,000	600	132	No hypoxia
10,000	523	110	Mild symptoms of hypoxia start appearing
15,000	400	90	Moderate hypoxia develops with following symptoms: – Reduction in visual acuity – Effects on mental functions: – Improper judgment and – Feeling of over confidence
20,000	349	73	Severe hypoxia appears with cardiorespiratory symptoms such as – Increase in heart rate and cardiac output – Increase in respiratory rate and respiratory minute volume This is the highest level for permanent inhabitants
25,000	250	62	This is the critical altitude for survival – Hypoxia becomes severe – Breathing oxygen becomes essential
29,628	235	49	This is the height of Mount Everest
30,000	226	47	Symptoms become severe even with oxygen
50,000	87	18	Hypoxia becomes more severe even with pure oxygen

CHANGES IN THE BODY AT HIGH ALTITUDE:

- When a person is exposed to high altitude, particularly by rapid ascent, the various systems in the body cannot cope with lowered oxygen tension and effects of hypoxia start. Besides hypoxia, some other factors are also responsible for the changes in functions of the body at high altitude.

Factors Affecting Physiological Functions at High Altitude

1. Hypoxia
2. Expansion of gases
3. Fall in atmospheric temperature
4. Light rays.

EFFECT OF HYPOXIA:

- Some of the important acute effects of hypoxia in the unacclimatized person breathing air, beginning at an altitude of about 12,000 feet, are drowsiness, lassitude, mental and muscle fatigue, sometimes headache, occasionally nausea, and sometimes euphoria. These effects progress to a stage of twitchings or seizures above 18,000 feet and, above 23,000 feet in the unacclimatized person, end in coma, followed shortly thereafter by death.

EFFECTS OF EXPANSION OF GASES ON THE BODY:

- Volume of gases increases when the barometric pressure is reduced. So at high altitude, due to the decreased barometric pressure, volume of all gases increases in atmospheric air, as well as in the body.
- At the sea level with atmospheric pressure of 760 mm Hg, if the volume of gas is 1 liter, at the height of 18,000 feet (where atmospheric pressure is 379 mm Hg), it becomes 2 liter. And it becomes 3 liter, at the height of 30,000 feet (where atmospheric pressure is 226 mm Hg).
- Expansion of gases in GI tract causes painful distention of stomach and intestine. It is minimized by supporting the abdomen with a belt or by evacuation of the gases. Expansion of gases also destroys the alveoli. During very rapid ascent from sea level to over 30,000 feet height, the gases evolve as bubbles, particularly nitrogen, resulting in decompression sickness

EFFECTS OF REDUCED ATMOSPHERIC TEMPERATURE:

- Environmental temperature falls gradually at high altitudes. The temperature decreases to about 0°C at the height of 10,000 feet. It becomes –22°C at the height of 20,000 feet. At the altitude of 40,000 feet, the temperature falls to –44°C. Injury due to cold or frostbite occurs if the body is not adequately protected by warm clothing.

EFFECTS OF LIGHT RAYS:

- Skin becomes susceptible for injury due to many harmful rays like ultraviolet rays of sunlight. Moreover, the sunrays reflected by the snow might injure the retina of the eye, if it is not protected with suitable tinted glasses. Severity of all these effects depends upon the speed at which one ascends in high altitude. The effects are comparatively milder or moderate in slow ascent and are severe in rapid ascent.

ACCLIMATIZATION OF LOW PO2:

- Acclimatization refers to the adaptations or the adjustments by the body in high altitude. While staying at high altitudes for several days to several weeks, a person slowly gets adapted or adjusted to the low oxygen tension, so that hypoxic effects are reduced. It enables the person to ascent further.
- Various changes that take place during acclimatization help the body to cope with adverse effects of hypoxia at high altitude. Following changes occur in the body during acclimatization:

CHANGES DURING ACCLIMATIZATION:

1. Changes in Blood: During acclimatization, RBC count increases and packed cell volume rises from normal value of 45% to about 59%. Hemoglobin content in the blood rises from 15 g% to 20 g%. So, the oxygen carrying capacity of the blood is increased. Thus, more oxygen can be carried to tissues, in spite of hypoxia. Increase in packed cell volume and hemoglobin content is due to erythropoietin actions. Increase in RBC count, packed cell volume and hemoglobin content is due to erythropoietin, that is released from juxtaglomerular apparatus of kidney.
2. Changes in Cardiovascular System: Overall activity of cardiovascular system is increased in high altitude. There is an increase in rate and force of contraction of the heart and cardiac output. Vascularity in the body is increased due to vasodilatation induced by hypoxia. So, blood flow to vital organs such as heart, brain, muscles, etc. increases.

3. Changes in Respiratory System

i. Pulmonary ventilation: Pulmonary ventilation increases up to 65%. This is the immediate compensation for hypoxia in high altitude and this alone helps the person to ascend several thousand feet. Increase in pulmonary ventilation is due to the stimulation of chemoreceptors .

ii. Pulmonary hypertension: Increased cardiac output increases the pulmonary blood flow that leads to pulmonary hypertension. It is very common even in persons acclimatized to high altitude. In some of these persons, pulmonary hypertension is associated with right ventricular hypertrophy.

iii. Diffusing capacity of gases: Due to increased pulmonary blood flow and increased ventilation, diffusing capacity of gases increases in alveoli. It enables more diffusion of oxygen in blood.

4. Changes in Tissues: Both in human beings and animals residing at high altitudes permanently, the cellular oxidative enzymes involved in metabolic reactions are more than the inhabitants at sea level. Even when a sea level inhabitant stays at high altitude for certain period, the amount of oxidative enzymes is not increased. So, the elevation in the amount of oxidative enzymes occurs only in fully acclimatized persons. An increase in the number of mitochondria is observed in these persons.

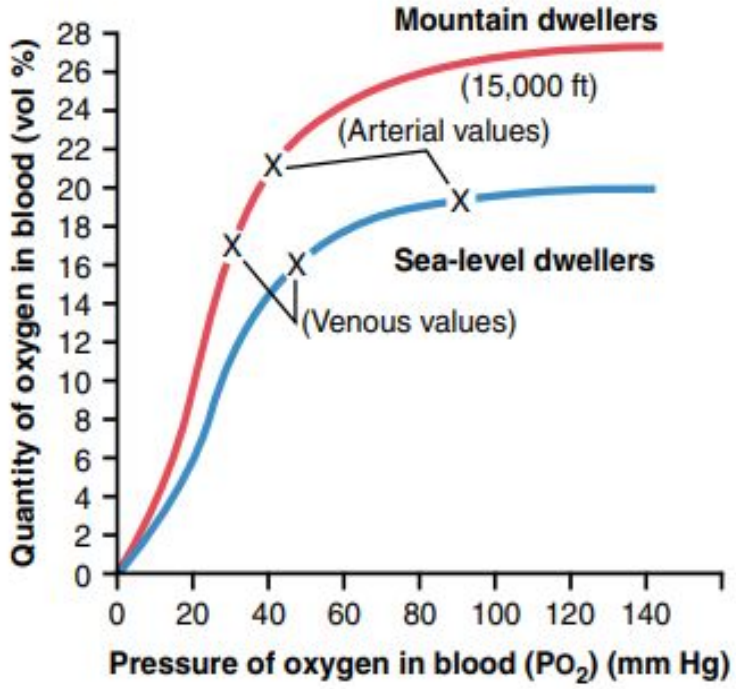
NATURAL ACCLIMATIZATION OF NATIVE PEOPLE LIVING AT HIGH ALTITUDE:

Anatomical Adaptations:

- **Respiratory:** High-altitude natives have significantly **increased chest sizes** relative to reduced body sizes, creating a high ratio of ventilatory capacity to body mass.
- **Cardiovascular:** Continuous exposure to high cardiac output from birth results in **considerably larger hearts** compared to lowlanders.

Oxygen Delivery & Efficiency:

- **Hemoglobin Content:** Despite an exceptionally low arterial oxygen pressure PO2 **of 40 mmHg**), high-altitude natives carry a greater total quantity of arterial oxygen than sea-level natives due to a **higher concentration of hemoglobin**.
- **Tissue Transport:** Oxygen delivery is highly effective; the venous PO2 of high-altitude natives drops by **only 15 mmHg** compared to lowlanders, demonstrating highly optimized tissue oxygenation.



	Work capacity (percent of normal)
Unacclimatized	50
Acclimatized for 2 months	68
Native living at 13,200 feet but working at 17,000 feet	87

MOUNTAIN SICKNESS:

- Mountain sickness is the condition characterized by adverse effects of hypoxia at high altitude. It is commonly developed in persons going to high altitude for the first time. It occurs within a day in these persons, before they get acclimatized to the altitude.
- SYMPTOMS: In mountain sickness, the symptoms occur mostly in digestive system, cardiovascular system, respiratory system and nervous system. Symptoms of mountain sickness are:

1. Digestive System: Loss of appetite, nausea and vomiting occur because of expansion of gases in GI tract.

2. Cardiovascular System: Heart rate and force of contraction of heart increases.

3. Respiratory System: Pulmonary blood pressure increases due to increased blood flow. Blood flow increases because of vasodilatation induced by hypoxia. Increased pulmonary blood pressure results in pulmonary edema, which cause breathlessness.

4. Nervous System: Symptoms occurring in nervous system are headache, depression, disorientation, irritability, lack of sleep, weakness and fatigue. These symptoms are developed because of cerebral edema. Sudden exposure to hypoxia in high altitude causes vasodilatation in brain. Autoregulation mechanism of cerebral blood flow fails to cope with hypoxia. It leads to an increased capillary pressure and leakage of fluid from capillaries into the brain tissues.

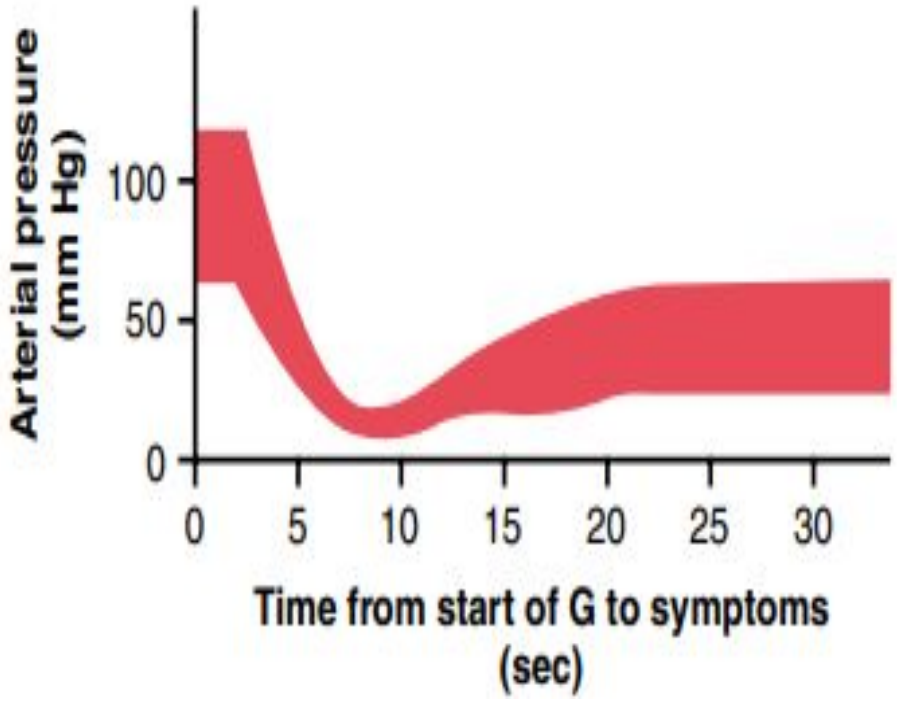
TREATMENT: Symptoms of mountain sickness disappear by breathing oxygen.

CHRONIC MOUNTAIN SICKNESS

- AVIATION PHYSIOLOGY:
- ACCELERATIVE FORCE: Acceleration means change in velocity. Flying straight in horizontal plane with constant velocity has minimum effects on the body. However, changes in velocity produce severe physiological effects. Accelerative forces are developed in the flight during linear, radial or centripetal and angular acceleration.
- **Gravitational force develops accelerative force** and is measured in G units, where 1 G equals normal body weight on Earth. G units increase during acceleration (positive G) and decrease during deceleration (negative G), with sudden changes in speed or direction centrifuging passengers in the opposite direction. Both the magnitude of these G forces and the direction they enter the body cause significant physiological changes during flight.

EFFECTS OF GRAVITATIONAL FORCES ON THE BODY:

- **Positive G forces push blood into the lower body and abdomen**, which drastically reduces cardiac output. This drops blood pressure and cuts off the oxygen supply (hypoxia) to the head, starving both the brain and eyes of necessary perfusion
- Grayout: **Grayout is a graying of vision** caused by reduced blood flow to the hypoxia-sensitive retina, serving as an early warning of decreased cerebral perfusion before physical impairment occurs.
- **Blackout: Hypoxia completely cuts vision** while consciousness and movement remain, signaling an imminent risk of passing out.
- **Loss of consciousness: Forces above 5 G cause fainting** and potential convulsions for 15 seconds, requiring 10 to 15 minutes of reorientation upon recovery.
- **Fracture of bones: Forces around 20 G fracture bones**, particularly the spine, even in a sitting position.



- **Hyperemia: Forces of –4 to –6 G push blood to the head,** causing facial flushing, mild headaches, and temporary brain edema that the body can safely withstand.
- **Redout and headache: Forces of –15 to –20 G engorge head and eye vessels,** turning vision red, though pooling cerebrospinal fluid cushions and protects brain vessels from rupturing.
- **Loss of consciousness: High negative G forces elevate chest and neck vessel pressure,** triggering bradycardia and severe blood stagnation that ultimately cause a blackout.

PREVENTION OF EFFECTS OF G FORCES ON THE BODY: Body can be protected from the effects of G forces, particularly positive G by the following methods:

1. By Using Abdominal Belts: Pooling of blood in the abdominal blood vessels is prevented by using abdominal belt and leaning forward while sitting in the aircraft. This procedure postpones gray out or blackout.

2. By Using Anti-G Suit: AntiG suit exerts a positive pressure on lower limbs and abdomen and prevents the pooling of blood in lower part of the body.

EFFECTS OF TRAVEL BY SPACECRAFT:

- **Space Environment:** **Human survival is challenged by differences in atmosphere, radiation, and gravity**, though spacecraft environmental systems and pressurized suits maintain Earth-like conditions.
- **Acceleration/Deceleration:** **Intense symptoms occur only during blast-off and landing**, with minimal forces during transit since spacecraft cannot make rapid directional changes.
- **Cardiovascular & Kidneys:** **Microgravity shifts fluids to the head and upper body**, triggering the kidneys to excrete water and electrolytes; this decreases blood volume and initially enlarges then shrinks the heart, causing dizziness without stimulating thirst.

- **Blood: Plasma volume and red blood cell counts drop**, a condition known as space anemia.
- **Immune System: Space travel suppresses** the body's immune system.
- **Musculoskeletal: Floating eliminates gravitational resistance**, causing muscle atrophy, loss of strength/endurance, bone weakening via increased osteoclastic activity, and calcium excretion in urine.
- **Space Motion Sickness: Abnormal vestibular stimulation and fluid shifts cause temporary motion sickness** (nausea, vomiting, headache, malaise) that typically disappears after two to three days.

DEEP SEA PHYSIOLOGY:

High Barometric Pressure Dynamics:

- **Deep-sea or mine environments present high barometric pressures** that compress internal organs and decrease gas volumes, increasing by 1 atmosphere for every 33 feet (10 meters) of depth.

Necessity of Compressed Air:

- **Divers must breathe compressed air** to equalize the intense external pressure acting against their thoracic wall and abdomen.
- **Nitrogen Narcosis Mechanism**
- **Breathing pressurized air forces normally inert nitrogen into body tissues,** inducing an altered mental state of stupor, lethargy, and suppressed sensations that closely mimics alcohol intoxication.

TABLE 129.1: Barometric pressure and its effects at different depth

Depth (feet)	Atmospheric pressure (mm Hg)	Effects on the subject
Sea level	1	–
33	2	–
66	3	–
100	4	Symptoms of nitrogen narcosis appear
133	5	Lack of concentration Becomes jovial and careless
166	6	Starts feeling drowsy
200	7	Feels fatigued, weak and careless
233	8	Looses power of judgment Unable to do skilled work
266	9	Becomes unconscious
Barometric pressure: 1 atmosphere = 760 mm Hg		

Mechanism:

- **Nitrogen dissolves easily in fat under high barometric pressure**, escaping blood vessels to pool inside neural membranes. This dissolved nitrogen acts as an anesthetic agent, suppressing neuronal excitability for as long as the person remains at depth.

Symptoms:

- **120 feet: Carelessness and joviality** develop, alongside a failure to comprehend surrounding dangers.
- **150 to 200 feet:** **Drowsiness** sets in.
- **200 to 250 feet:** **Extreme fatigue and weakness** manifest, accompanied by a complete loss of concentration, judgment, and skilled motor skills.
- **Beyond 250 feet:** **Unconsciousness** occurs.

Oxygen Toxicity:

- **Breathing hyperbaric oxygen at extreme depths raises blood PO2** to levels that overwhelm tissue antioxidant defenses, generating toxic free radicals that oxidize cellular lipids and enzymes.
- **CNS Toxicity:** High pressures quickly trigger sudden **grand mal seizures and coma**, often preceded by tunnel vision and facial twitching.
- **Pulmonary Toxicity:** Prolonged exposure damages respiratory tissue, causing **airway congestion, alveolar collapse, and pulmonary edema**.

Carbon Dioxide Toxicity:

- **Inadequate ventilation: due to dense compressed air or equipment failure causes CO2 buildup**, severely depressing the central nervous system.
- **Respiratory Acidosis:** Alveolar PCO2 levels exceeding 80 mmHg suppress the brain stem's respiratory center and throw blood pH out of balance.
- **Symptoms:** This toxic buildup triggers **severe lethargy, confusion, panic, and blackouts** while actively accelerating the onset of oxygen toxicity and nitrogen narcosis.

Decompression Sickness:

- **Rapid ascent to normal atmospheric pressure** from high-pressure environments like the deep sea causes decompression sickness (also known as dysbarism, compressed air sickness, caisson disease, bends, or diver's palsy).

Cause:

- **High barometric pressure compresses body gases** and reduces their overall volume. While oxygen is consumed by metabolic tissues and carbon dioxide is expired, inert nitrogen remains trapped inside body compartments.

Prevention:

- **Mixing helium with oxygen prevents nitrogen narcosis** by substituting nitrogen and diluting oxygen concentrations during deep dives. While helium can cause mild nausea and dizziness, its side effects are far less severe than nitrogen narcosis. Avoid alcohol for 24 hours prior to diving, maintain equipment, and limit work effort.

Treatment:

- **Returning to a depth of 60 feet completely resolves symptoms** without any lasting hangover effects, making further medical intervention unnecessary unless the diver lost consciousness.

THANK YOU